



Quick Commerce Data Analytics

Executive insights for fulfillment performance, operational efficiency, and growth opportunities.

Core Business Challenge

Quick commerce operations are constrained by four factors that weaken fulfillment and limit revenue.

Stockouts

Frequent out-of-stock events

Demand Volatility

Limited forecasting visibility

Inventory Imbalance

Stock levels do not match demand

Replenishment Gaps

Slow reorder cycles



Project Objectives

Five analytical goals guiding this project — from demand forecasting to replenishment intelligence.

01

Forecast demand

02

Track performance KPIs

03

Optimize inventory levels

04

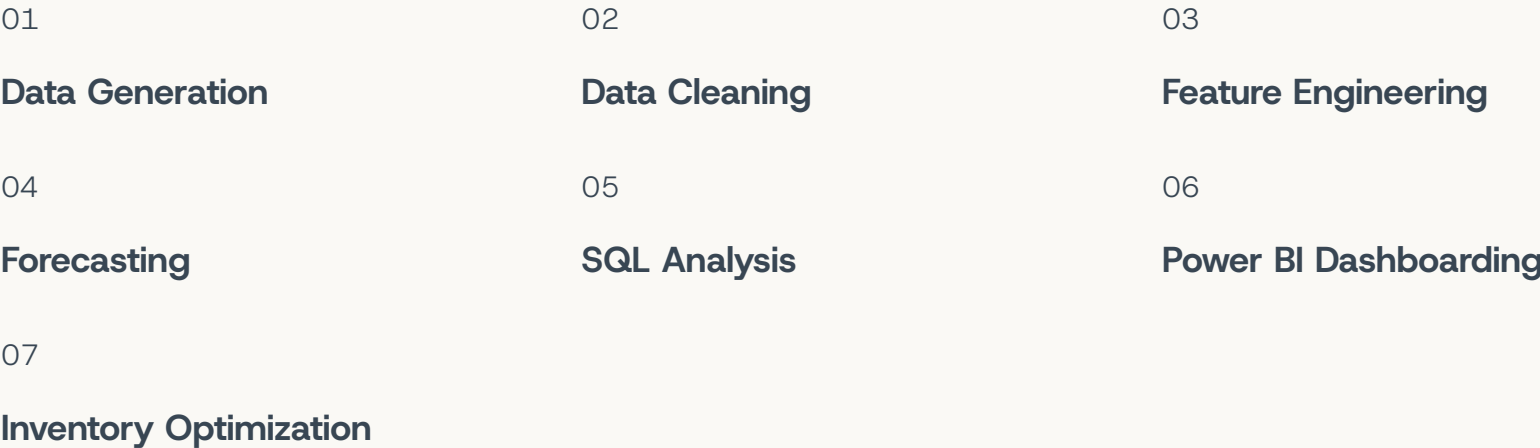
Improve replenishment planning

05

Reduce stockout risk

Project Workflow

A clean, vertical process flow showing the seven project stages from data creation to inventory decisions.





Dataset Overview

Simulated retail operations data with weather-driven demand signals.

3 Stores

Multi-location coverage

18 Products

Across core categories

Weather Inputs

Demand signal modeling

Performance KPIs

Fill rate, stockouts, forecast accuracy

Data Cleaning & Feature Engineering

Python preprocessing improved dataset reliability and forecasting readiness.



Fill Rate Creation

Calculated fulfilled demand as a product-store ratio.



Stockout Validation

Flagged out-of-stock events for risk analysis.



Demand Gap Analysis

Generated demand-inventory gap signals by SKU.



Weekend Demand

Identified weekend-driven demand spikes.



Forecast Features

Built rolling 7-day inputs for predictive modeling.

FORECASTING

Forecasting Analysis

A rolling 7-day forecasting model used historical product-store demand behavior to estimate future operational demand trends.

84.38%

Forecast Accuracy

Rolling 7-day model
performance

41.92

Average Forecast

Mean predicted demand
per SKU

6.08

Demand Variability

Operational demand
fluctuation index

18

High-Risk SKUs

Products with elevated
forecast error



Forecast instability was primarily concentrated among fast-moving operational SKUs.

Operational KPI Analysis

409,879

Total Demand

59.5%

SKU-Level Stockout Exposure

363,823

Fulfilled Quantity

28,349

Lost Sales

87.6%

Fill Rate

Category & Product Risk Analysis



Key Risk Findings

Category-level analysis identified clear inventory pressure across Essentials, Beverages, and Dairy. Beverages recorded the **lowest fill rate at 83.37%**, while Essentials drove the **highest lost sales volume**.

Top Risk Products

- ☐ Paneer – 347
- Oil – 342
- Juice – 339

Other High-Risk Products

- Rice – 337
- Buns – 332

SQL-Based Operational Analysis

Our SQL-based analytical framework provides deep insights into key operational areas, improving decision-making and performance optimization.

Inventory Movement Tracking

Real-time stock monitoring across stores

Demand Pattern Recognition

Temporal demand analysis by product and store

Stockout Event Correlation

Root-cause analysis of out-of-stock events

Replenishment Efficiency

Reorder cycle and lead time optimization

Category Performance Benchmarking

Cross-category inventory and fulfillment metrics

Dashboard Storytelling

Four Power BI dashboards delivered end-to-end operational intelligence across KPIs, forecasting, inventory risk, and optimization.



Operational KPI Dashboard

Tracks demand, stockout exposure, fulfillment, and lost sales across stores and categories.



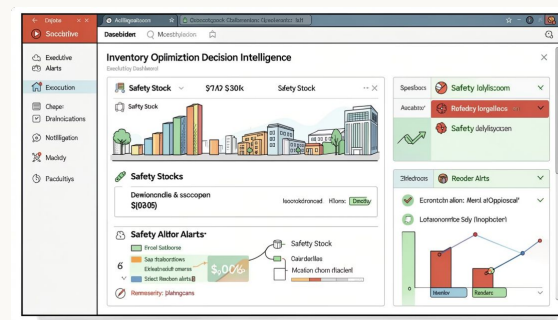
Forecasting Dashboard

Highlights demand movement, forecast accuracy, weekend behavior, and variability to support planning.



Inventory Risk Dashboard

Surfaces high-risk products, category pressure, and stockout exposure by store.



Inventory Optimization Dashboard

Supports replenishment planning, safety stock decisions, reorder intelligence, and demand simulation.

Clean dashboard grid with consistent preview framing and concise explanations for each BI view.

Inventory Optimization & Decision Intelligence

Advanced optimization techniques improve inventory efficiency and reduce operational risk.

Optimization Metrics

10,000

Reorder Alerts

Automated replenishment triggers

47.4%

Optimization Score

Inventory efficiency level

23.94%

Safety Stock

Coverage

Buffer stock adequacy

491.9K

Scenario Demand

Projected demand volume

Key Optimization Insights

- Safety stock modeling identified critical buffers for high-risk SKUs
- Reorder point optimization reduced stockout exposure and excess inventory
- Demand simulations tested replenishment strategies across seasonal patterns
- Allocation rules improved inventory distribution across stores

Key Findings & Recommendations

Key Business Findings

- Stockout exposure is elevated at 59.54%, indicating material replenishment risk
- Forecast variability is concentrated in fast-moving SKUs
- Inventory optimization upside exists across all product categories
- Weather has a moderate but measurable impact on demand patterns

Recommendations

- Increase safety stock for high-risk SKUs: Paneer, Oil, Juice, Rice, and Buns
- Increase replenishment frequency for Essentials and Beverages
- Adopt adaptive forecasting for products with higher volatility
- Track weekend demand spikes separately to improve planning accuracy
- Reduce lost sales with proactive reorder planning and tighter thresholds



Conclusion

This project transformed operational retail data into actionable business intelligence using Python, SQL, and Power BI, improving visibility into demand behavior, inventory performance, forecasting variability, and replenishment planning.